MIL-S-45885(PA)
25 February 1976
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCH, ROTARY

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for four types of Rotary Switches used in high-reliability circuit applications in the Lance warhead section.

1.2 Classification. This specification covers rotary switches of the following:

S1	12 positions, two poles, make-before-break.
S2	12 positions, two poles, make-before-break.
S3	6 positions, two poles, make-before-break.
S4	12 positions, nine poles, make-before-break.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-W-16878/D	- Wire, Electrical, Insulated, High Temperature, Type EE.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking for U.S. Military Property.
MIL-STD-456	- Electronic Parts, Date and Source Coding for.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

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MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

9236777	- Switch, Rotary (S1 and S2).
9236778	- Switch, Rotary (S3).
9236779	- Switch, Rotary (S4).
EL-9236777	- Equipment List.
EL-9236778	- Equipment List.
EL-9236779	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switches shall comply with all requirements specified on the drawing (dwgs) 9236777, 9236778 and 9236779, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall affix serial numbers to switches for control purposes. Unless otherwise specified on the contract or order, the contractor may assign serial numbers at his discretion.

3.5 Hookup wire. All hookup wire shall comply with MIL-W-16878/D, Type EE.

3.6 Lacing. Lacing shall be made of polyimide and have no coating material.

3.7 Dissimilar metals. Dissimilar metals (see 6.6) shall not be used in combination unless suitably coated to prevent electrical corrosion.

3.8 Cadmium. Cadmium shall not be used in the fabrication of any part in the assembly.

3.9 Contact. The contact material shall be capable of complying with all the tests herein.

3.10 Soldering. Soldering shall be performed in accordance with MIL-STD-1460.

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3.11 Operating torque. When switches are tested in acccordance with 4.4.1, the torque required to rotate the shaft from each detent position to the next detent position shall be 8.0 ± 2.0 in-lbs.

3.12 Enclosure seal. The switch enclosure purged and filled with 90% nitrogen and 10% helium, shall have a leak rate of 1×10^{-5} cubic centimeters per second maximum at an external pressure of 10 micron when tested as specified in 4.4.2. Enclosures shall be of sufficient mechanical strength to withstand the normal abuse incurred in handling, transit, storage, and installation without causing malfunction or distortion of parts.

3.13 Circuit resistance. The resistance of the circuits as indicated in Table I shall not exceed 100 milliohms when tested as specified in 4.4.3.

3.14 Insulation resistance. The insulation resistance of each circuit to all other circuits and case shall be 100 megohms minimum when tested as specified in 4.4.4.

(a) All terminals to case.
(b) With switch in Position 1:

for 9236777 (S1 and S2)	- terminals 1 thru 6, 8, 9, 10; 15, 16, 17, 21, 22, 23, 28, 29, 32, 33, 35, 36, 37, individually, to all other terminals.
for 9236778 (S3)	- terminals 2, 3, 4, 9, 22; 36 and 37, individually, to all other terminals.
for 9236779 (S4)	- terminals 1 thru 5; 7 thru 25; 27 thru 30; 32 thru 57; 59, 60, 61, individually, to all other terminals, however, exclude terminal 3 to terminal 30.

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Table I

	9236777 (S1 and S2)	9236778 (S3)	9236779 (S4)		
Position 2	35-32	36-22	27-44*	4-8	28-39
	36-37	37-22	52-60	54-40	47-55
			35-30	36-3	15-41*
Position 3	35-33	36-9	27-51	4-14*	28-46
	36-28	37-4	52-50	54-49*	47-42
			35-3	36-3	15-41
Position 4	35-22	36-3	27-59	4-20	28-53
	36-21	37-2	52-34*	54-33	47-24*
			35-3	36-3	15-41
Position 5	35-15		27-61*	4-45	28-48
	36-9		52-18	54-25	47-23
			35-3	36-3	15-41*
Position 6	35-8		27-43	4-57	28-32
	36-4		52-10	54-5*	47-16
			35-3*	36-3	15-41*
Position 7	35-3		27-11	4-56	28-22
	36-2		52-3	54-3	47-9*
			35-38	36-17	15-41
Position 8	35-1		27-11	4-56	28-22
	36-6		52-2*	54-1	47-7
			35-38*	36-17	15-41
Position 9	35-5		27-11	4-56	28-22
	36-10		52-12	54-13	47-21*
			35-38	36-17	15-41
Position 10	35-17		27-11	4-56*	28-22
	36-16		52-19	54-29	47-37
			35-38	36-17*	15-41*
Position 11	35-23				
	36-29				

NOTE: When testing switch S4 as specified in 3.18.6, only the circuits denoted by (*) shall be tested.

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3.15 Chatter. Contact chatter shall not exceed 0.000010 seconds duration at a current of 100 ± 0.5 milliamperes (ma) when tested as specified in 4.4.5.

3.16 Operational load. Each individual contact shall be subjected to a resistive load current of 2.0 ± 0.2 amperes for 3 minutes minimum when tested as specified in 4.4.6. The switches shall comply with the requirements of 3.13 and 3.14 after testing.

3.17 Rotation. When switch assemblies are tested as specified in 4.4.7, there shall be no evidence of broken, loose, deformed or displaced parts. Uniform contact pressure shall be maintained and shall permit smooth electrical and mechanical control of the switch. Contacts shall be self-cleaning with no adjustments required during testing. The test switches shall comply with the requirements of 3.11 through 3.14 after testing.

3.18 Environmental testing. The switches shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switches shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.8. Tests shall be conducted in the order specified.

3.18.1 Low temperature. The switch shall withstand a temperature of -60 ± 5°F for 72 hours minimum without evidence of external damage (see 6.5) when tested as specified in 4.4.8.1. The switch shall comply with the requirements of 3.11 and 3.13 during testing and 3.11 through 3.13 after testing.

3.18.2 High temperature. The switch shall withstand a temperature of +165 ± 5°F for 72 hours minimum without evidence of external damage (see 6.5) when tested as specified in 4.4.8.2. The switch shall comply with the requirements of 3.11 and 3.13 during testing and 3.11 through 3.13 after testing.

3.18.3 Humidity and temperature. The switch shall withstand ten (10) 24-hour humidity-temperature cycles having a temperature from 165 ± 5°F to 80 ± 20°F and a relative humidity of 95 ± 5 percent without evidence of external damage (see 6.5) when tested as specified in 4.4.8.3. The switch shall comply with the requirements of 3.11 and 3.13 during testing and 3.11, 3.12 and 3.14 after testing.

3.18.4 Thermal shock. The switch shall withstand three (3) alternate exposures of 165 ± 5°F and -60 ± 5°F without evidence of external damage (see 6.5) when tested as specified in 4.4.8.4. The switch shall comply with the requirements of 3.11, 3.12 and 3.14 after testing.

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3.18.5 Mechanical shock. The switch shall withstand a force of 50 ± 5 gravity units (g's) peak amplitude, three (3) times in each direction along its three (3) mutually perpendicular axes (total 18 shocks), without evidence of external damage (see 6.5) when tested as specified in 4.4.8.5. The time duration of the half sine wave shock shall be 11 ± 1.0 milliseconds (ms). The switch shall comply with the requirements of 3.15 during testing and 3.11 through 3.14 after testing.

3.18.6 Vibration. The switch shall withstand vibration tests along each of its three (3) mutually perpendicular axes as specified in the Table below without evidence of external damage (see 6.5) when tested as specified in 4.4.8.6. The switch shall comply with the requirements of 3.15 during testing and 3.11, 3.12 and 3.13 after testing.

Vibration test schedule

a. 0.3 g^2/Hertz (Hz) power spectral density (19.0 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibels per octave so that the total frequency enveloped is 50 to 2000 Hz for 10 seconds. The tolerance level shall be + 3 decibels (Db) between 100 to 1000 Hz and ± 6 Db between 1000 to 2000 Hz.

b. 0.06 g^2/Hz power spectral density (8.5 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 12 Db so that the frequency enveloped is 50 to 2000 Hz for one (1) minute. The tolerance level shall be ± 3 Db between 100 to 2000 Hz.

3.18.7 Acceleration and spin. The switch shall withstand the loading imposed by the following simultaneous levels of acceleration for a period of three minutes minimum in each direction along each of its three mutually perpendicular axes when tested as specified in 4.4.8.7.

a. Constant acceleration of 30 g's (maximum).

b. Centripetal acceleration resulting from a constant rotational speed of 300 rpm with the longitudinal axis of the switch located 8 inches from the center of rotation. The switch shall comply with the requirements of 3.13 and 3.15 during testing and 3.13 after testing.

3.18.8 Transportation vibration. The switch shall withstand the environmental properties listed in Table II, in each direction along its three mutually perpendicular axes without evidence of external damage when tested as specified in 4.4.8.8. The switch shall comply with the requirements of 3.11 through 3.14 after testing.

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Table II

Vibration test schedule

Frequency (Hz)	Intensity Level Double Amplitude	Intensity Level Gravity Units	Sweeptime ± 1/2 minute -60°±5°F	Sweeptime ± 1/2 minute +165°±5°F
5 - 10	0.04 inch	---	9	9
10 - 15	0.04 inch	---	9	9
15 - 80	---	5 g's peak	9	9
80 - 115	0.015 inch	---	9	9
115 - 500	---	10 g's peak	9	9

3.18.9 Salt spray. The switch shall withstand the salt spray environment in accordance with MIL-E-5272, Procedure I when tested as specified in 4.4.8.9. The switch shall comply with the requirements of 3.11 after testing.

3.18.10 Fungus. The switch shall withstand the fungus environment in accordance with MIL-E-5272, Procedure I when tested as specified in 4.4.8.10. The switch shall comply with the requirements of 3.11 after testing.

3.19 Marking. Switches shall be marked in accordance with MIL-STD-130 with the following information:

(a) Manufacturer's name or trademark.

(b) Government dwg number, type designation, and serial number (see 3.4).

(c) Date code in accordance with MIL-STD-456.

(d) Switch terminals shall be marked as specified in the applicable detail specification or dwg.

The finish or any dimension of the item shall not be altered or damaged by application of marking information. The material used for marking shall not be subject to corrosion, fungus, flaking or similar adverse action which could prevent proper functioning of the switches.

3.20 Workmanship.

3.20.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.20.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.20.3 Burrs. Parts shall be free from burrs.

3.20.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.20.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.20.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 32 complete assemblies (see dwgs 9236777, 9236778 and 9236779) for first article testing to the testing facility designated in the invitation for bids or request for proposal. When the contractor is producing all four types on the same contract, the first article lot shall be composed of 4 S1's, 4 S2's, 8 S3's and 16 S4's.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

4.2.2.1 Group A inspection. Select sixteen (16) switches at random from the first article lot. However, if the lot is composed of the four types of switches, select a proportional quantity of each type to represent Group A and the sub-lots. Subject the switches to the test requirements in the order listed below:

Test Requirements

(a) Vibration (see 3.18.6)
(b) Rotation (see 3.17)
(c) Simultanious operation environments:
  (1) Sub-lot 1 - 8 units, stabilize at -65° ± 5°F (see 6.4), hold at this temperature for 48 hours minimum and test as specified in 4.4.7 and 4.4.6.
  (2) Sub-lot 2 - 8 units, test at +165° ± 5°F temperature, as specified in 4.4.8.6 and 4.4.6.

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4.2.2.2 Group B inspection. The remaining sixteen (16) switches of the first article lot shall be subjected to the test requirements in the order listed in Table IV. The evaluation checks specified during and after each test shall be deleted. Upon the completion of all tests in Table IV, each switch shall be subjected to test requirements 2 through 7 of Table III.

Table III

First article inspection and tests

Inspection or Test	Dest	Non-Dest	Units 1-32	Testing *Group A	Sequence **Group B
1. Classification of characteristics (see 4.3.2.1)		X	X		
2. Operating torque (see 3.11)		X	X		
3. Enclosure seal (see 3.12)		X	X		
4. Circuit resistance (see 3.13)		X	X		
5***Insulation resistance (see 3.14)		X	X		
6. Chatter (see 3.15)		X	X		
7. Operational load (see 3.16)		X	X		
8. Vibration (see 3.18.6)	X			X	
9. Rotation (see 3.17)	X			X	
10. Simultaneous operation environments (see 4.2.2.1)	X			X	
11. Environmental tests (see 4.2.2.2)	X				X

*See 4.2.2.1.
**See 4.2.2.2.
***The potential shall be applied for 60 seconds minimum on all initial tests and thereafter shall be tested in accordance with 4.4.4.

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Table IV

Test	Units 1 S1	2 S1	3 S2	4 S2	5 S3	6 S3	7 S3	8 S3	9 S4	10 S4	11 S4	12 S4	13 S4	14 S4	15 S4	16 S4
Low temp (see 3.18.1)				X			X	X		X		X	X	X	X	
High temp (see 3.18.2)			X			X	X		X		X	X	X	X		
Therm shock (see 3.18.4)		X			X	X		X		X	X	X	X			
Transp vib (see 3.18.8)	X			X	X		X		X	X	X	X				
Mech shock (see 3.18.5)			X	X		X		X	X	X	X				X	
Accel spin (see 3.18.7)		X	X		X		X	X	X	X				X		
Hum temp (see 3.18.3)	X	X		X		X	X	X	X				X			
Salt spray (see 3.18.9)	X		X		X	X	X	X				X				X
Fungus (see 3.18.10)		X		X	X	X	X				X			X		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch, rotary (see dwgs 9236777, 9236778 and 9236779).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Body outside dia (except 9236779) incorrect	Gage
102.	Body outside dia (except 9236779) incorrect	Gage
103.	Body overall length (except 9236779) incorrect	Gage
104.	Body overall length (only 9236779) incorrect	Gage
105.	Alignment pin length, max, incorrect	Gage
106.	Shaft outside diameter, incorrect	Gage
107.	Shaft flat, incorrect	Gage
108.	Shaft thru hole diam., incorrect	Gage
109.	Shaft extension from mounting surface incorrect	Gage
110.	Objective evidence relative to material missing or improper (see 3.1 and 3.5 through 3.10)	Visual
111.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packaging, packing and marking not in accordance with MIL-STD-1169 (MU) (see 5.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Operating torque	Major
b.	Enclosure seal	Major
c.	Circuit resistance	Major
d.	Insulation resistance	Major
e.	Low temperature	Major
f.	High temperature	Major
g.	Rotation (test length of 15 rotations)	Major
h.	Low temperature	Major
i.	High temperature	Major
j.	Vibration	Major
k.	Rotation	Major

Of these tests, (h) through (k) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Operating torque, enclosure seal, circuit resistance and insulation resistance. Each switch shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.11 through 3.14, the item shall be rejected and removed from the lot.

4.3.3.2 Low temperature, high temperature, rotation, and vibration. A sample shall be selected at random in accordance with Table VII and subjected to the test requirements of 4.3.3.2.1 and 4.3.3.2.2. When the contractor is producing the four types of switches on the same contract, the sample lot make-up shall be as follows:

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Switch Type	Percent of Sample Lot Size
S1	25
S2	25
S3	25
S4	25

4.3.3.2.1 Destructive lot sampling inspection. Select two each S1, S2, S3, and S4 switches from the random sample described in 4.3.3.2 and subject each switch to the test requirements of (h) through (k).

4.3.3.2.2 Non-destructive lot sampling inspection. The remainder switches from the random sample described in 4.3.3.2 shall each be subjected to the test requirements of (e) through (g). The switches subjected to (e) through (g) only may be returned to the lot.

Table VII

Sample size for lot testing.

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-9236777, 9236778 and 9236779).

When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Operating torque. The switch shall be mounted on an approved fixture. The torque required to rotate the switch from one position to the next shall be measured and recorded for all positions in the clockwise and counter-clockwise directions. The torque shall be within the limits specified in 3.11.

4.4.2 Enclosure seal. The switch shall be placed in an approved fixture and utilizing a helium mass spectrometer leak detector measure the switch enclosure leak rate as specified in 3.12.

4.4.3 Circuit resistance. The switch shall be tested on approved resistance measuring equipment with each circuit carrying an applied current of 100 ± 1 milliamperes (ma) while being tested to determine compliance with 3.13. Separate voltage and current hookup wiring shall be used at each terminal in order not to include the effect of hookup wire on measurement values. Activation of switches for purpose of improving contact surfaces prior to actual measurement of resistance shall not be permitted.

4.4.4 Insulation resistance. The insulation resistance shall be measured on approved equipment at a potential of 500 ± 50 volts direct current (vdc) with current limited to 5 ma. The potential shall be applied for 5 seconds minimum to 60 seconds maximum while testing to determine compliance with 3.14.

4.4.5 Chatter. The switch shall be measured on approved equipment which will record contact chatter to determine compliance with 3.15.

4.4.6 Operational load. The applied current through each contact shall be measured on approved current measuring equipment, which will indicate current and time duration of application for each circuit to determine compliance with 3.16.

4.4.7 Continuous rotation. The switch assemblies shall be mounted on an approved fixture in the normal operating position and rotated in an unenergized condition for a test length of 500 rotations to determine compliance with 3.17. The rotary speed between positions shall be 1.0 seconds minimum.

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4.4.8 Environmental testing (see 3.18).

4.4.8.1 Low temperature. The switch shall be placed in an approved chamber and the temperature lowered to -60 ± 5°F for a soak period of 72 hours minimum and evaluated for compliance to 3.13. The switch shall then be removed from the chamber and allowed to return to 80 ± 20°F, without the use of forced heating, and tested for compliance with 3.18.1.

4.4.8.2 High temperature. The switch shall be placed in an approved chamber and the temperature raised to +165 ± 5°F for a soak period of 72 hours minimum and evaluated for compliance to 3.13. The switch shall then be removed from the chamber and allowed to return to 80 ± 20°F, without the use of forced cooling, and tested for compliance with 3.18.2.

4.4.8.3 Humidity and temperature. The switch shall be placed into an approved chamber and subjected to the test requirements in 3.18.3. Each 24-hour humidity-temperature cycle shall be as follows:

(a) Uniformly raise the internal chamber temperature to +165 ± 5°F and relative humidity to 95 percent over a period of 2 hours.

(b) Maintain conditions of step (a) for 6 hours minimum.

(c) Maintain 95 percent relative humidity minimum and uniformly lower chamber temperature over a period of 16 hours to 80 ± 20°F.

(d) Repeat steps (a), (b) and (c) for 10 cycles total. At the completion of the 10th cycle the switch shall be removed from the chamber and dried by applying filtered compressed air for 5 minutes or until dry. After the switch has dried, it shall be visually examined for external damage (see 6.5) and tested to determine compliance with 3.18.3.

4.4.8.4 Thermal shock. The switch shall be placed in an approved temperature chamber stabilized at +165 ± 5°F (see 6.4) and the switch shall remain at this temperature for a minimum period of four hours. At the conclusion of this period, and within five minutes, the switch shall be transferred to another approved chamber stabilized at -60° ± 5°F (see 6.4). The switch shall remain at this temperature for a minimum period of four hours. This constitutes one cycle. The number of complete cycles shall be three. At the conclusion of the third cycle, the switch shall be removed form the test chamber, returned to standard temperature, and within a period of one hour test for compliance with 3.18.4.

4.4.8.5 Mechanical shock. The switch shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed on a shock producing device and tested to the levels and tests as specified in 3.18.5.

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4.4.8.6 Vibration. The switch shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed on a vibrator and vibrated to the levels and tests as specified in 3.18.6.

4.4.8.7 Acceleration and spin. The switch shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed on a centrifuge and accelerated to the levels and tests as specified in 3.18.7.

4.4.8.8 Transportation vibration. The switch shall be mounted securely on a fixture capable of being positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed in a suitable temperature vibration chamber set at -65 + 5°F for 8 hours minimum. The switch shall then be vibrated as specified in 3.18.8. The switch shall then be prepared and tested as above at a temperature of +165 + 5°F. Remove the switch from the chamber, allow to return to room temperature, visually examine for external damage, and test for compliance with 3.18.8.

4.4.8.9 Salt spray. The switch shall be subjected to the test procedure for salt spray in accordance with MIL-E-5272 for a minimum period of 50 hours. After completion, the switch shall comply with 3.18.9.

4.4.8.10 Fungus. The switch shall be subjected to the test procedure for fungus in accordance with MIL-E-5272. After completion, the switch shall comply with 3.18.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switches shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The switches shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The switches shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 External damage. Characteristics such as obliterated markings, damaged finish exposing base metal, rust and cracks that are detectable by visual examination, shall be considered cause for rejection.

6.6 Dissimilar metals. Dissimilar metals are defined in Table VIII.

Table VIII

Grouping of metals

Group I	Group II	Group III	Group IV
Magnesium alloys (most anodic)	Aluminum	Zinc	Copper and its alloys
	Aluminum alloys	Cadmium	Nickel and its alloys
	Zinc	Steel	Chromium
	Cadmium	Lead	Stainless steel
	Tin	Tin	Gold
	Stainless steel	Stainless steel	Silver (most cathodic)

(a) Contact between a member of any one group and another member of the same group is considered as contact of similar metals. Contact between a member of one group and a member of any other group is considered as contact of dissimilar metals, except for zinc, tin and cadmium as listed in Groups II and III, and for stainless steel as listed in Groups II, III and IV.

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(b) Unless specifically indicated by the procuring activity, all other metals are considered dissimilar with respect to each other and with respect to any of the materials listed in Table VIII.

(c) Care should be exercised in using aluminum alloys against each other or against differing materials.

(d) Where reference is made to a metal in a particular group, the metal applies to the metal on the surface of the part; that is, zinc means zinc castings as well as zinc electro plate, zinc hot dip, or zinc metal spray.

(e) Different metals in contact, even though similar, should be employed in assemblies in such a manner that the smaller part is cathodic or protected and the larger part is anodic or corroded, if any corrosion takes place.

6.7 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2029 dated 9 March 1971, with Amendment 1, dated 10 October 1972.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A066

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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